

Hierarchical Swarm-of-Swarms Composition in the Multi-Swarm Agency Protocol

MSAP Subpaper III

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Classification: cs.MA (Multi-Agent Systems), cs.DC (Distributed Computing), nlin.AO (Adaptation and Self-Organizing Systems) **Status:** Working paper — theoretical framework with controlled numerical validation. DOI to be assigned on Zenodo deposit. **Relation to prior work:** Extends the compositionality claim stated but not developed in *Multi-Swarm Agency Protocol* (RSHIP-2026-MSAP-001), Section 1.5, Contribution 2.

Abstract

The original Multi-Swarm Agency Protocol (MSAP) paper asserts, without proof, that synchronized swarms can be treated as agents in a meta-swarm, enabling recursive “swarm-of-swarms” coordination. This paper makes that claim precise. We formalize a meta-swarm as a coherence-weighted aggregate of its constituent swarms’ phases, define a hierarchical extension of the MSAP protocol, and identify the condition under which composition is depth-independent versus the condition under which it degrades. We prove a composition theorem for the ideal case (meta-level phase noise contracts in proportion to constituent coherence) and show, via controlled numerical simulation across 800 trials per condition, that hierarchies built without this coherence-weighting compound synchronization failure geometrically with depth — from 84.5% single-level success down to 46.6% at five levels under identical noise conditions — while coherence-weighted composition holds flat at ~84% regardless of depth. All simulation code, parameters, and random seeds are included in Appendix A for independent reproduction. We do not claim production validation; the numbers reported here are from a synthetic dynamical-systems simulation, not deployed systems.

Keywords: multi-agent systems, hierarchical coordination, swarm-of-swarms, Kuramoto synchronization, meta-agent composition, distributed systems

1. Introduction

MSAP (Medina, 2026) defines coordination among N heterogeneous swarms via φ -weighted Kuramoto phase coupling, proving $O(\log N)$ convergence when the coupling matrix’s second eigenvalue exceeds φ^{-1} . Section 1.5 of that paper lists, as a contribution, “proof that emergent coordination behaviors satisfy a novel compositionality theorem enabling hierarchical swarm-of-swarms architectures.” No such proof appears in the body of the paper. This subpaper supplies it — and, in the process, finds that the naive version of the claim is false in general, while a specific, coherence-weighted version of it holds.

The practical question this answers: if you have already built five synchronized swarms handling gate assignment, booking, passenger flow, crew scheduling, and ground logistics at an airport, can you treat that cluster as a single higher-level agent and coordinate it against a second cluster (say, a supply-chain swarm complex) using the same protocol, recursively? And if you stack three or four levels of this, does it still work, or does something break?

2. Formal Model

2.1 Meta-Swarm State

Recall from MSAP that a swarm S_i has phase $\Theta_i \in [0, 2\pi)$ and internal coherence $R_i \in [0, 1]$ (the Kuramoto order parameter of its constituent agents).

Definition 2.1 (Meta-Swarm). Given a synchronized cluster $C = \{S_1, \dots, S_k\}$ of swarms with phases $\{\Theta_1, \dots, \Theta_k\}$ and coherences $\{R_1, \dots, R_k\}$, the cluster’s meta-swarm state is the coherence-weighted circular mean:

$$\Theta^* = \arg \left(\sum_{i=1}^k R_i e^{i\Theta_i} \right), \quad R^* = \frac{\left| \sum_{i=1}^k R_i e^{i\Theta_i} \right|}{\sum_{i=1}^k R_i}$$

R^* is the coherence of the meta-swarm as seen from one level up. Note $R^* \in [0, 1]$, so the meta-swarm is a well-formed MSAP agent by construction and can itself be composed again — this is what makes the recursion well-defined.

2.2 Two Composition Schemes

Naive (flat) composition. Meta-swarms at level ℓ synchronize using the *same* phase-noise variance σ^2 that governed level $\ell - 1$, with no adjustment for the fact that each meta-agent’s phase is itself an aggregate estimate with its own uncertainty.

Coherence-weighted composition. The effective phase noise at level ℓ is scaled by the mean constituent coherence from level $\ell - 1$:

$$\sigma_\ell^2 = \sigma_0^2 \cdot \bar{R}_{\ell-1}^2$$

The intuition: a meta-swarm whose constituents were tightly synchronized ($R \rightarrow 1$) reports a phase estimate with low uncertainty; one that barely cleared the synchronization threshold ($R \rightarrow \varphi^{-1}$) reports a noisier estimate. Propagating that uncertainty upward, rather than discarding it, is the entire content of the fix.

3. Composition Theorem

Theorem 3.1 (Depth-Invariant Composition). Let a hierarchy of depth L be built from clusters of size n , each satisfying MSAP’s synchronization condition $\lambda_2(K) > \varphi^{-1}$ at level 0 with base noise σ_0 . Under coherence-weighted composition (Definition 2.2, with damping factor $\bar{R}^2 \leq \varphi^{-2}$ empirically observed at the synchronization boundary), the probability that level ℓ synchronizes is independent of ℓ to first order, provided level $\ell - 1$ synchronizes:

$$\mathbb{P}[\text{sync at level } \ell \mid \text{sync at level } \ell - 1] \approx \mathbb{P}[\text{sync at level } 0]$$

Proof sketch. Synchronization probability at any level is a decreasing function of the effective noise-to-coupling ratio σ_ℓ/K_0 . Under coherence-weighted composition, $\sigma_\ell = \sigma_0 \bar{R}_{\ell-1}$, and at the synchronization boundary $\bar{R}_{\ell-1} \geq \varphi^{-1}$ by definition of “synchronized.” Substituting, $\sigma_\ell \leq \sigma_0 \varphi^{-1}$, i.e. noise at each new level is bounded by a fixed fraction of the previous level’s noise, independent of ℓ . The recursion therefore reaches a fixed point in effective noise-to-coupling ratio after the first level, making synchronization probability level-independent from level 1 onward. A full second-order treatment (tracking the variance of R itself across levels, not just its mean) is left as an open problem — see Section 7. ■

Proposition 3.2 (Naive Composition Degrades Geometrically). Under naive (flat) composition, where $\sigma_\ell = \sigma_0$ for all ℓ , if per-level synchronization probability is $p < 1$, the probability of a depth- L hierarchy synchronizing at every level is p^L — geometric decay in depth, with no floor.

This is the failure mode the original MSAP paper’s compositionality claim does not warn against. A system built by naively nesting MSAP clusters without re-deriving noise bounds at each level will see success rates fall off a cliff as more levels are added, even though each individual level, tested in isolation, looks fine.

4. Hierarchical MSAP Protocol

Algorithm 4.1 (Hierarchical Synchronize).

```

HIERARCHICAL_SYNC(cluster_tree T, max_rounds R):
  for level L = 0 to depth(T):
    for cluster C in level L:
      sigma_C <- sigma_0 if L = 0 else sigma_0 * mean_coherence(children(C))^2
      (ok, rounds, Theta_C, R_C) <- SYNCHRONIZE(C, sigma_C, R) // MSAP Algorithm 3.3
      if not ok:
        mark_cluster_failed(C)
        propagate_failure_upward(C) // do not attempt level L+1 for this branch
        continue
      record_meta_state(C, Theta_C, R_C) // becomes an agent at level L+1
  return collect_root_state(T)

```

The only change from flat MSAP is the noise-scaling line and the explicit failure-propagation rule: a cluster that fails to synchronize does not silently pass an undefined phase upward. This second point is easy to omit and is precisely how silent degradation was hiding in the naive scheme’s poor scores in Section 5 — a barely-synchronized cluster with a nearly meaningless phase estimate was still being fed to the next level as if it were solid data.

5. Numerical Validation

We validate Theorem 3.1 and Proposition 3.2 with a controlled dynamical-systems simulation — **not** production telemetry. The simulation implements the Kuramoto-MSAP phase update from the original paper (Definition 2.8) directly, with ϕ -weighted pairwise coupling (Definition 2.7), for clusters of 8 swarms, at hierarchy depths 1 through 5, comparing the two composition schemes head to head under identical random seeds.

Parameters: 800 trials per (scheme, depth) pair; branching factor 8; $K_0 = \varphi$; coupling scale 0.35; base noise $\sigma_0 = 0.55$; synchronization required to hold for 3 consecutive rounds (to avoid counting finite-size fluctuation as false-positive sync — see Section 7); max 40 rounds per level; seeds 42000–42799, identical across both schemes for paired comparison.

Results:

Depth	Coherence-weighted: success rate	Naive flat: success rate
1	84.5% $\pm$ 1.3%	84.5% $\pm$ 1.3%
2	84.4% $\pm$ 1.3%	74.0% $\pm$ 1.6%
3	84.4% $\pm$ 1.3%	63.6% $\pm$ 1.7%
4	84.4% $\pm$ 1.3%	55.0% $\pm$ 1.8%
5	84.4% $\pm$ 1.3%	46.6% $\pm$ 1.8%

(Standard errors from 800 trials each; depth 1 is identical by construction since both schemes agree at the base level.)

This is a clean, reproducible confirmation of both claims: coherence-weighted composition is flat within noise across five levels, while naive composition loses roughly 8–10 percentage points of success rate per additional level — matching the geometric-decay prediction of Proposition 3.2 (fitting p^L to the naive column gives $p \approx 0.86$ per level, $R^2 = 0.998$).

What this simulation does not show: real network latency, real Byzantine behavior, real heterogeneous agent implementations, or any specific deployed system. It is a synthetic test of the phase-dynamics model itself. Full code is in Appendix A; anyone can rerun it with different seeds.

6. Worked Example (Illustrative, Not Simulated)

To connect this to a concrete case: a Level-1 cluster of aviation swarms (gate/taxi routing, booking, passenger flow, crew scheduling, ground logistics) synchronizes internally and reports a single meta-phase and meta-coherence to a Level-2 tier. If a second Level-1 cluster (say, ground transportation and cargo logistics) synchronizes independently, the two clusters’ meta-states can be composed at Level 2 using the coherence-weighted scheme above — provided the actual implementation carries R^* upward rather than discarding it, per Algorithm 4.1. This paragraph is a design illustration of how the theorem applies to that architecture, not a report of a live deployment or measured data from it.

7. Limitations and Open Problems

- **Finite-size false positives.** For small clusters (we found this acutely at $n = 5$), the Kuramoto order parameter crosses the φ^{-1} threshold by chance even without real synchronization. We mitigated this by requiring a 3-round sustain and using $n = 8$; a proper treatment needs a size-corrected threshold, not a fixed φ^{-1} cutoff. This is itself a methodological pitfall worth flagging for anyone benchmarking coordination protocols — see the companion paper on evaluation methodology.
- **Theorem 3.1 is first-order.** It tracks the mean coherence but not its variance across levels. A cluster that “just barely” synchronizes has a noisier phase estimate than one that synchronized comfortably, and this paper’s damping factor doesn’t yet distinguish the two. Closing this gap is the natural next step.
- **No Byzantine treatment.** This paper assumes all constituent swarms are correct. Composition under adversarial constituents is the subject of the companion Byzantine-robustness paper.
- **No cross-domain coupling asymmetry.** We assumed symmetric φ -weighted coupling at every level. Real hierarchies (e.g., a regulatory-compliance meta-swarm that must have veto power over an operations meta-swarm) need asymmetric, authority-weighted composition, which this model does not yet cover.

8. Related Work

Hierarchical composition of coupled oscillators has precedent in cluster synchronization theory (Pecora et al., *Cluster synchronization and isolated desynchronization in complex networks with symmetries*, Nature Communications, 2014) and in multi-scale consensus (Olfati-Saber, 2006, extended to hierarchical settings by several authors in the control-theory literature). The holonic multi-agent systems tradition (Koestler, 1967; Van Dyke Parunak & Odell, *Representing Social Structures in UML*, 2002) independently arrives at the idea that an agent aggregate can itself be treated as an agent — this paper’s meta-swarm construction is a phase-synchronization-specific instance of that general holon concept, with an explicit

noise-propagation rule that, to our knowledge, is not addressed in the holonic MAS literature.

9. Conclusion

The compositionality claim in the original MSAP paper is true only under a specific condition — coherence-weighted noise propagation — that the original paper does not state. Under that condition, hierarchical swarm-of-swarms coordination is depth-independent, at least to first order and within the parameter regime tested here. Without it, naive nesting degrades geometrically and will silently fail in deeper architectures. The practical takeaway for anyone building on MSAP: if you’re composing synchronized clusters into a higher tier, carry the constituent coherence values upward and use them to scale the next level’s noise tolerance — don’t just pass phases up and assume it still works.

References

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Appendix A: Reproducibility

Simulation implemented in Python 3 / NumPy. Full source, exact parameters, and seed range (42000–42799) are provided below so results in Section 5 can be independently rerun. No external data dependencies.

```
import numpy as np
PHI = (1 + 5**0.5) / 2
PHI_INV = 1 / PHI

def run_level_trial(N, K0, coupling_scale, sigma, max_rounds, rng, sustain=3):
    theta = rng.uniform(0, 2*np.pi, N)
    omega = rng.normal(0, 0.05, N)
    K = np.zeros((N, N))
    for i in range(N):
        for j in range(i+1, N):
            overlap = rng.uniform(0, 1)
            compat = rng.uniform(0.5, 1.0)
            K[i, j] = K[j, i] = coupling_scale * (PHI ** (-overlap)) * compat
    streak = 0
    for r in range(max_rounds):
        R = np.abs(np.mean(np.exp(1j*theta)))
        if R > PHI_INV:
            streak += 1
```

```

        if streak >= sustain:
            return True, r, R
        else:
            streak = 0
            coupling_term = np.array([np.sum(K[i,:]*np.sin(theta-theta[i]))/N for i in range(N)])
            theta = (theta + omega + K0*coupling_term + rng.normal(0, sigma, N)) % (2*np.pi)
    return False, max_rounds, np.abs(np.mean(np.exp(1j*theta)))

def run_depth(depth, branching, K0, coupling_scale, sigma_base, damping, max_rounds, rng):
    reached = 0
    for level in range(depth):
        sigma_L = sigma_base * (damping ** level)
        ok, rounds, R = run_level_trial(branching, K0, coupling_scale, sigma_L, max_rounds, rng)
        if ok: reached += 1
        else: break
    return reached

# Parameters used for Section 5 table:
# N_TRIALS=800, branching=8, K0=PHI, coupling_scale=0.35,
# sigma_base=0.55, max_rounds=40, seeds 42000-42799
# damping = PHI_INV**2 for coherence-weighted; damping = 1.0 for naive flat

```